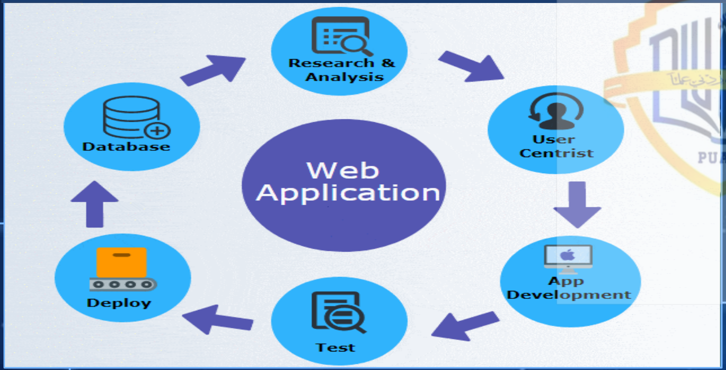
Course Title: Web Technologies
Course Code: DI 322
BS.IT 5TH Semester

Slides Made By **AREEJ ALI**

Let's Start

Table of contents



01

Introduction

What is a Web Applications?

Basic Definition:
A **web application** is a program that runs on a computer with a web server.

A **web application** (or **web app**) is application software that runs on a web server, unlike computer-based software programs that are run locally on the operating system (OS) of the device. Web applications are accessed by the user through a web browser with an active network connection. These applications are programmed using a client–server model—the user ("client") is provided services through an off-site server that is hosted by a third-party.

Examples of commonly-used web applications include:

- web-mail,
- online retail sales,
- online banking,
- and online auctions.

So what's a web server?

A web server is computer software and underlying hardware that accepts requests and send responses via HTTP or its secure variant HTTPS, the network protocols created to distribute web contents (web pages, etc.) to client user agents.

What does this mean? It means the client sends requests to the server and the server responds:

If the server always sends back the same data in the form of HTML, CSS, images, videos, and similar resources, without any user interaction, we call the app a **web site**.

If there is user interaction (mouse clicks, tapping and swiping, phone shaking, keypresses, etc.) that affect the execution of the program, we have a **web app**.

If the server only produces raw data (generally in text or JSON), then we speak of a **web service**.

Examples

Web Site

Strawberry Pop-Tart Blow-Torches is a **web site**. It’s just HTML with images. Last updated in 1994. Looks great.

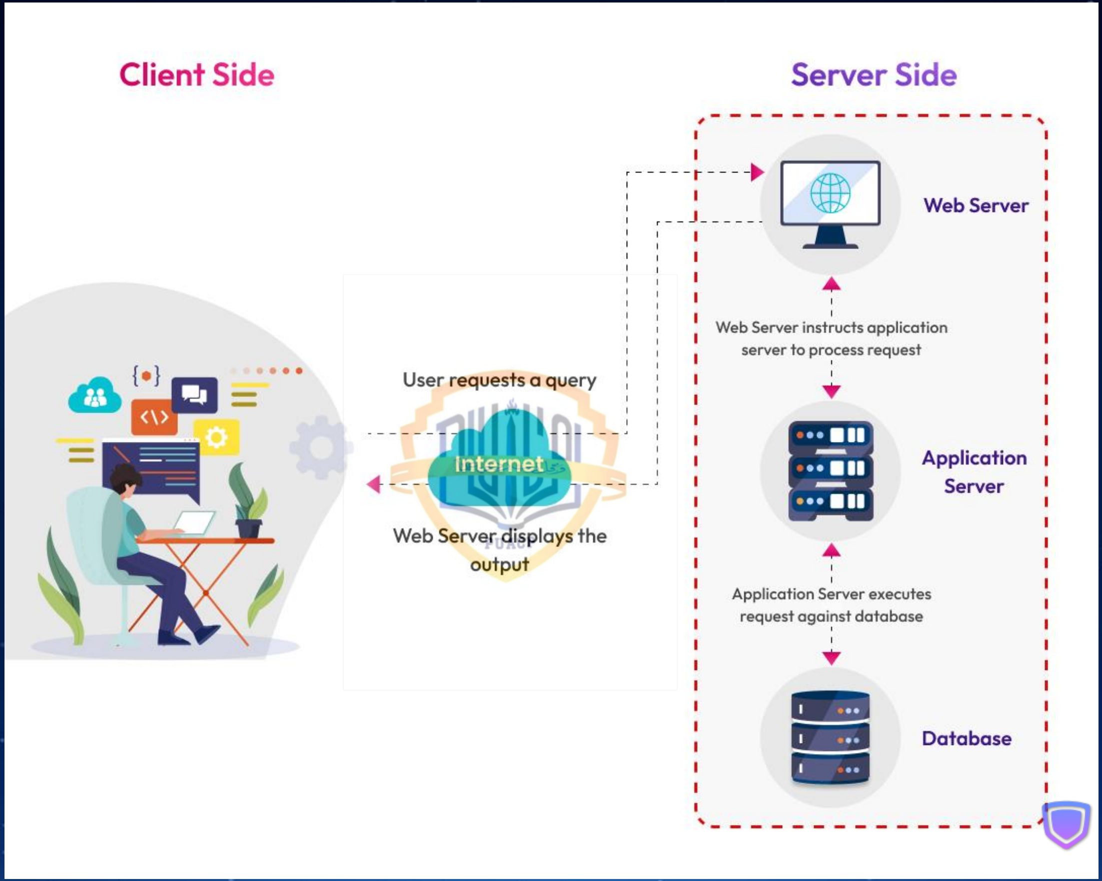
Web App

Gmail is a **web app**. All users need is a web browser. They login, create and organize filters, read messages, reply, forward, send, and delete, and logout. Messages exist in a data store on the server, as does all the code to generate pages. Of course the "pages" include a fair number of scripts that the browser knows how to execute, but these scripts are kept on the server and downloaded on demand.

Web Service

Enter **https://pokeapi.co/api/v2/pokemon/ditto** into your browser’s address bar. You got back pure data.

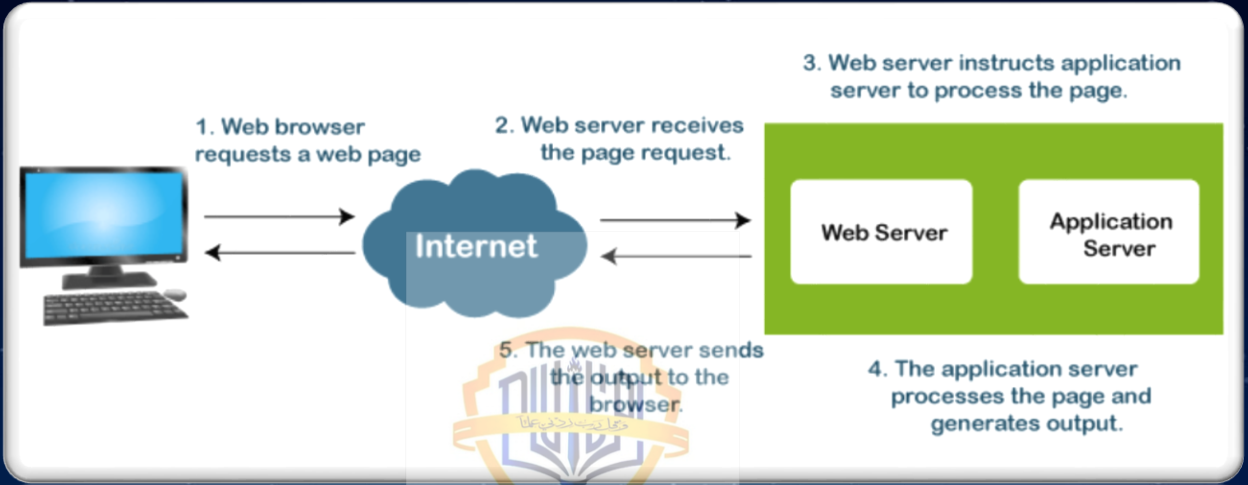
How does a web- application work?



How does a web- application work?

- Web applications allow users to access them through browsers such as Google Chrome, Opera, Firefox, or Safari rather than downloading the application. A web application requires only three components: a web server, an application server, and a database.
- A web server receives the requirements from the client, an application server processes the received request, and the data is saved in the database.
- **Step 1:** User triggers a request on the internet via a web browser.
- **Step 2:** The web server receives the request from the browser and instructs the **web application** server to execute the order.
- **Step 3:** The web application server fulfills the command by generating results by processing the database
- **Step 4:** The outcome is shared from the web application server to the web server.
- **Step 5:** The web server displays the received information on the screen for the requested user via desktop or mobile device.
- On the client side (front-end), the most commonly used are JavaScript, Cascading Style Scripts (CSS) and HTML5 for coding purposes. On the server side (back-end), Java, Ruby and PHP are some of the most popular languages.

The Flow of the Web Application



- In general, a user sends a request to the web-server using web browsers such as **Google Chrome, Microsoft Edge, Firefox**, etc over the **internet**.
- Then, the request is forwarded to the appropriate web **application server** by the **web-server**.
- Web application server performs the requested operations/ tasks like **processing the database, querying the databases; produces** the result of the requested data.
- The obtained result is sent to the web-server by the web application server along with the requested data/information or processed data.
- The web server responds to the user with the requested or processed data/information and provides the result to the user's screen .

What is Web Application Development?

Web application development entails creating an application using only a web browser, with only client-side and server-side programming required.
The creation process begins with the developer setting goals to solve a particular issue. The user interface is designed with the solution in mind.
Information requests from the customer are received from the user interface, therefore, development is designed based on receiving and responding to the information.
The web development team has to incorporate the following into their design:

- The application is compatible with both Android and iOS.
- Build a user-friendly and interactive user interface.

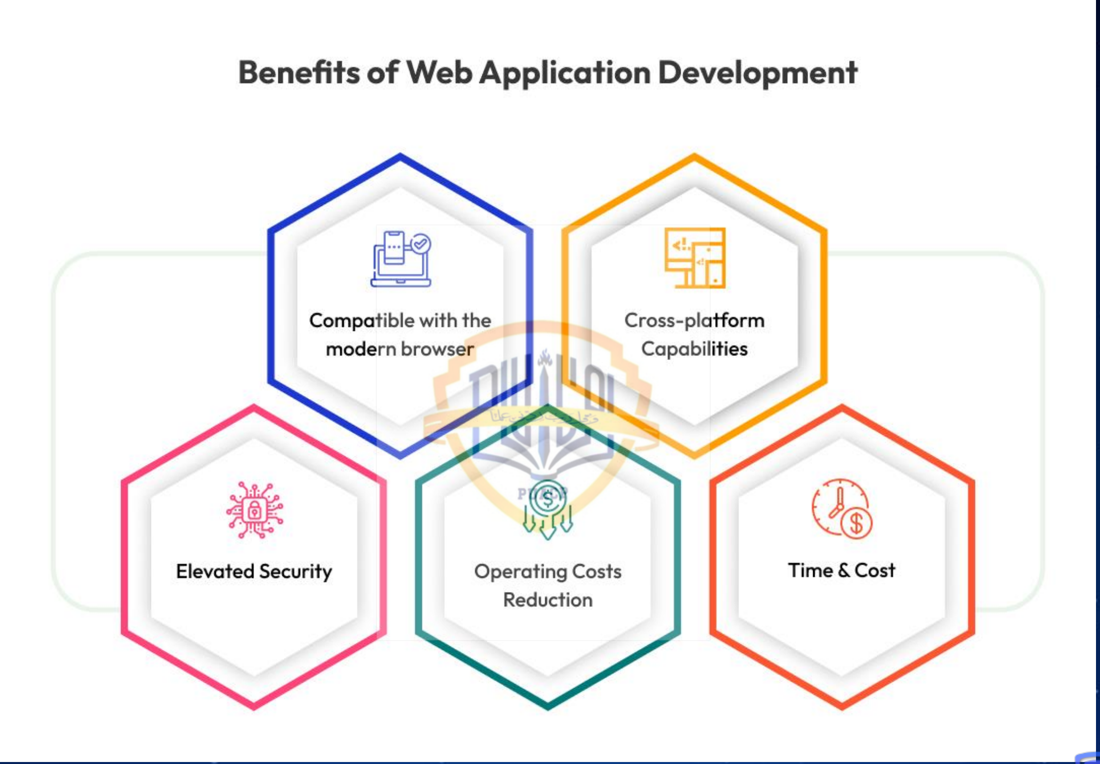
What are the different types of web applications?

Static Web Application	As the name suggests, in a static web application, there is no relationship built between the user and the server. It directly displays the content to the user without processing any data on the server side. This simple type of web application provides access to basic animated elements or videos. On the flip side, they are difficult to update or modify. Their usage is not interactive and is meant to provide information to the users without receiving any input from them. HTML, CSS, and JavaScript were used to create this page. Noted examples are Siteleaf and Netlify.
Dynamic Web Application	Like the previous application, there is no interaction between the user and the server. However, the dynamic web application is versatile in terms of flexibility and interactiveness. It responds to the user's needs in real time and gives them the freedom to post a comment, fill out a form, search for information, and so forth. On the flip side, they are more involved and complex in structure. They usually deploy a content management system to modify the content presented to end users. Technical tools: ASP.NET or PHP, were used to create this page. Noted examples are Facebook and Spotify.
E-commerce Web Application	An E-commerce web application is a commercial entity promoting the buying or selling of goods online. To be classified for e-commerce the web application needs to incorporate core features It distinguishes itself from dynamic web apps by incorporating core features such as an electronic payment gate, transaction integrations, a management panel for administrators (for managing orders, payments, updating, and deleting products), and users' cabinets. Technical tools: ASP.NET or PHP were used to create this page. Noted examples are Amazon, eBay, Walmart, Swiggy.
CMS Web Application	A content management system (CMS) allows users to produce, maintain and alter the content on the website. The person making alterations to the site requires no prior web programming or mark-up languages. Their usage is seen in personal blogs, corporate blogs and media outlets. Noted examples are WordPress, Drupal, and Joomla.
Single-Page Application	Single-Page Applications dynamically rewrite the current web page with new data, removing the need to reload the page to display new content. It is built using both front-end and back-end technology and provides a great experience for the end users. On the flip side, it is a complex type of web application. A notable example is Gmail.
Progressive Web Apps	Progressive Web Apps are the most recent and widely used web application innovation. It incorporates the best of both worlds, the features of a website and mobile app and gives a superior user experience in comparison to other types of web applications. It provides the user with options for installation, offline functionality, push notifications, and app access without an internet connection.

What is the difference between a web application and a website?

Differences	Web applications	Website
Authentication	Mandatory.	No requirement for information websites.
Precompilation	Required before deploying.	Not required.
Integration	With other software it is a complex and cumbersome process.	It is hassle-free.
Functionalities	Complicated and numerous.	Simpler.
User-engagement	Dynamic and Interactive.	Static and provide a single-path informational feed.

Benefits of Web Application Development?



THANK YOU!!!!!!

TOPIC CMPLETED